EXPERIMENT 1

EXPT NO: 1 VERIFICATION OF KIRCHHOFF'S LAWS

AIM:

a) To verify Kirchhoff's Voltage Law (KVL) for the given circuit.

b) To verify Kirchhoff's Current Law (KCL) for the given circuits.

APPARATUS REQUIRED:

S.no	Components	Range	Quantity
1	Resistor	1kΩ	3
2	Voltmeter (DC)	0-30V	3
3	Ammeter (DC)	(0-200)mA	3
4	Bread Board	-	1
5	Regulated Power Supply	(0-30)V	1
6	Connecting wires	As required	

THEORY:

KVL: Kirchhoff's voltage law states that the sum of the voltage differences around any closed loop in a circuit must be zero. A loop in a circuit is any path that ends at the same point at which it starts.

KCL: Kirchhoff's Current Law (KCL) Kirchhoff's Current Law states that the algebraic sum of the currents entering and leaving a node is equal to zero. By convention, currents entering the node are positive, and those leaving a node are negative.

PROCEDURE:

a) KVL:

1.Open Proteus 8.13 and create a new project.

2.Select and place a 12V DC Source,three resistors(R1,R2,R3), and four voltmeters.

3.Set the required values for the voltage source and resistors.

4.Connect the three resistors with the DC source in series to form a closed circuit.

5.Connect three voltmeters in parallel across the individual resistors and the fourth across the entire resistor network.

6.Run the simulation and note the voltage readings.

7.Verify KVL using the equation:

$V = V1+V2+V3$

b) KCL:

1.Open Proteus 8.13 and create a new project.

2.Select and place a 12V DC Source , four resistors(R1,R2,R3,R4),four ammeters.

3.Connect R1 in series with the source and form three branches using R2,R3 and R4.

4.Connect the ammeters in series with the main circuit and each branch to measure the currents.

5.Set the required values.

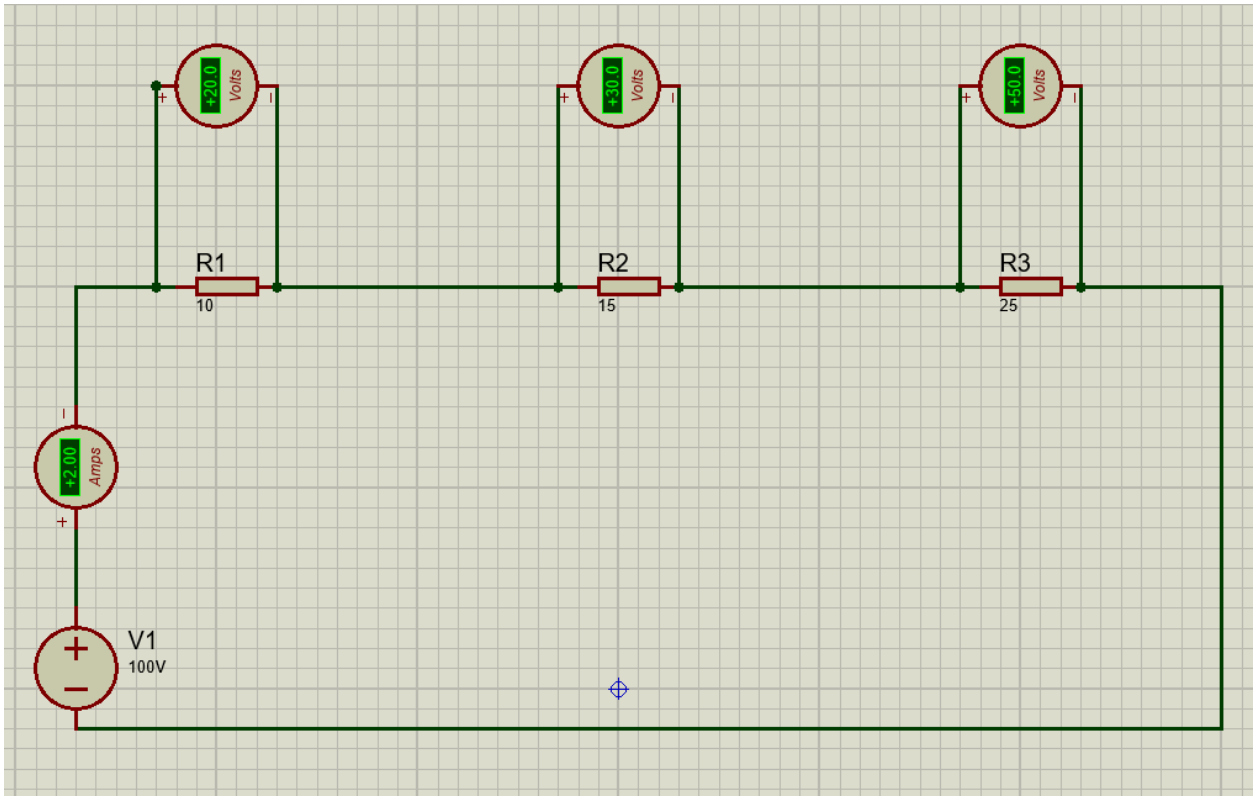
6.Run the simulation and note the current readings.

7.Verify KCL using the equation:

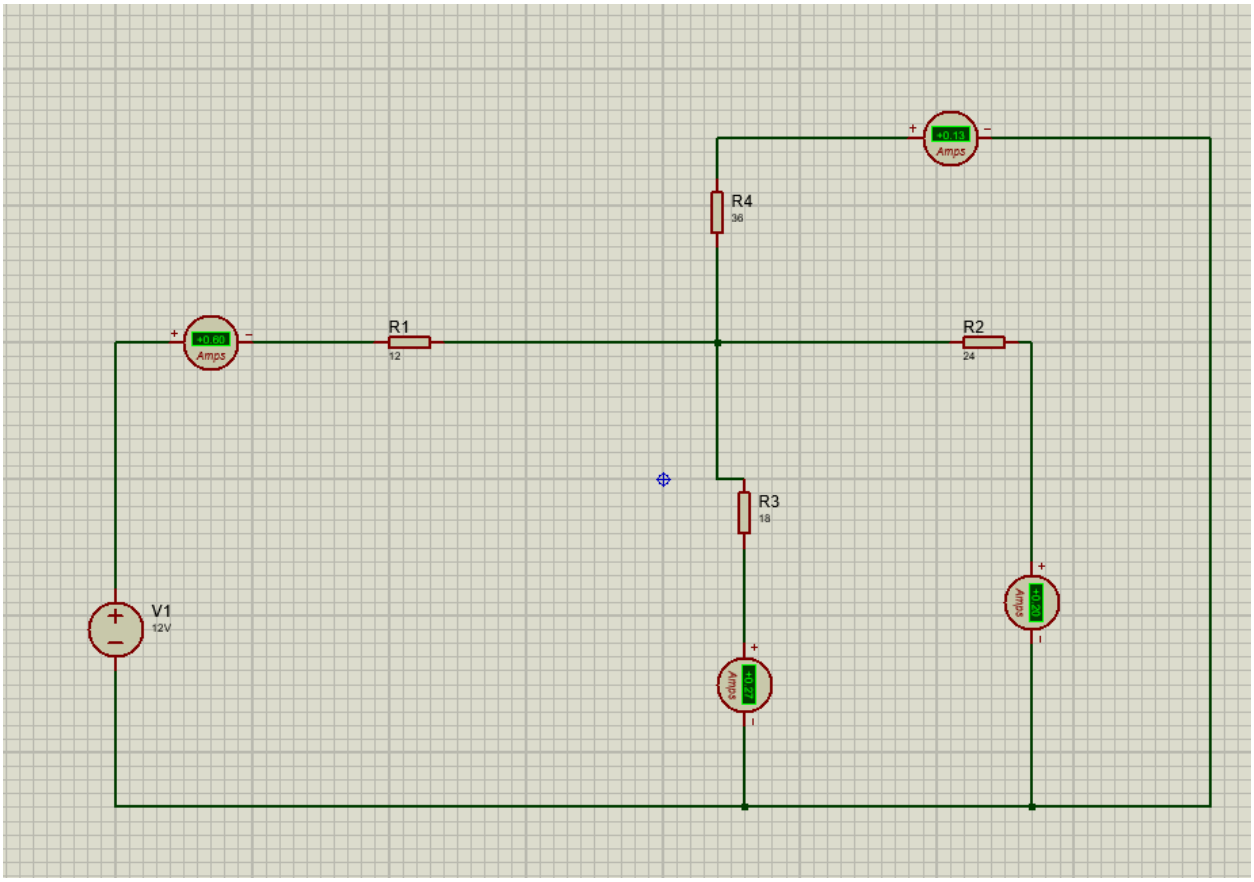
$I = I1+I2+I3$

CIRCUIT DIAGRAM:

a) KVL:



b) KCL:



CALCULATIONS:

a) KVL:

1) KVL:

$V = 100V$

$R_1 = 10\Omega$

$R_2 = 15\Omega$

$R_3 = 25\Omega$

Total Resistance:

$$R_T = R_1 + R_2 + R_3$$

$$= 10 + 15 + 25 = 50\Omega$$

Total Current:

$$I_T = \frac{V}{R_T} = \frac{100}{50} = 2A$$

Voltage drops:

$$V_1 = IR_1 = 2 \times 10 = 20\ V$$

$$V_2 = IR_2 = 2 \times 15 = 30\ V$$

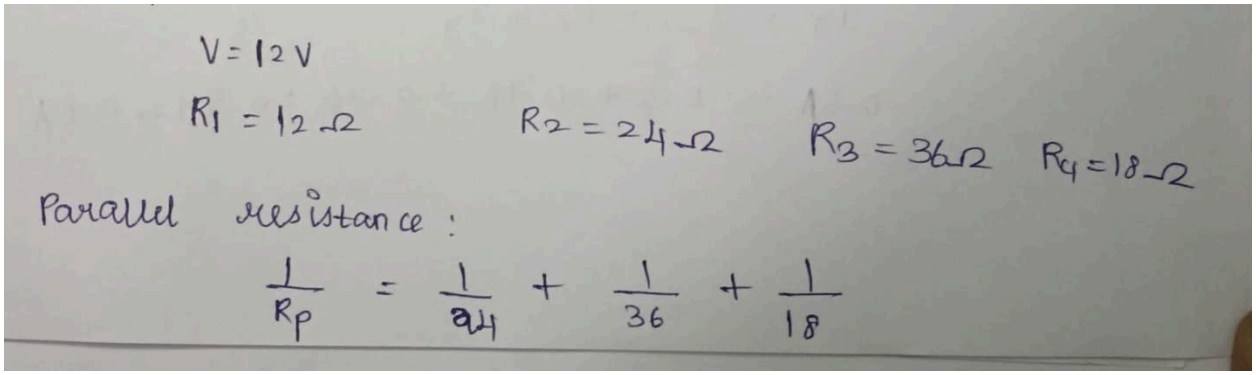
$$V_3 = IR_3 = 2 \times 25 = 50\ V$$

$$V = V_1 + V_2 + V_3$$

$$= 20V + 30V + 50V = 100V$$

$$100V = 100V$$

b) KCL:


V = 12 V
R1 = 12 Ω
R2 = 24 Ω
R3 = 36 Ω
R4 = 18 Ω
Parallel resistance :
1/Rp = 1/24 + 1/36 + 1/18

$$= \frac{2+3+4}{12} = \frac{9}{12} = \frac{1}{8}$$

$$R_P = 8\,\Omega$$

Total Resistance:

$$R_T = R_1 + R_P$$

$$= 12 + 8 = 20\,\Omega$$

Total current:

$$I_T = \frac{V}{R_T} = \frac{12}{20} = 0.6A$$

Voltage across the parallel network:

$$V_P = I_T \times R_P$$

$$= 0.6 \times 8 = 4.8V$$

Current through each resistor:

$$I_2 = \frac{4.8}{24} = 0.2A$$

$$I_3 = \frac{4.8}{36} = 0.13A$$

$$I_4 = \frac{4.8}{18} = 0.26A$$

Verification:

$$I_T = I_2 + I_3 + I_4$$

$$0.6A = 0.2 + 0.13 + 0.26 = 0.59 \simeq 0.6A$$

TABULATION:

a) KVL:

a) KVL :

Quantity	Calculated Value
Supply voltage	100V
Circuit current	2A
V_1	20V
V_2	30V
V_3	50V
Sum of voltage drops	100V

b) KCL:

b) KCL:

current	Theoretical value (A)	Simulated value (A)
I_1	0.6A	0.6A
I_2	0.2A	0.2A
I_3	0.13A	0.13A
I_4	0.26A	0.26A
Sum of outgoing current	0.6A	0.6A

RESULT:

Thus, for the given circuit, Kirchhoff's Laws (a) KVL and (b) KCL are proved.